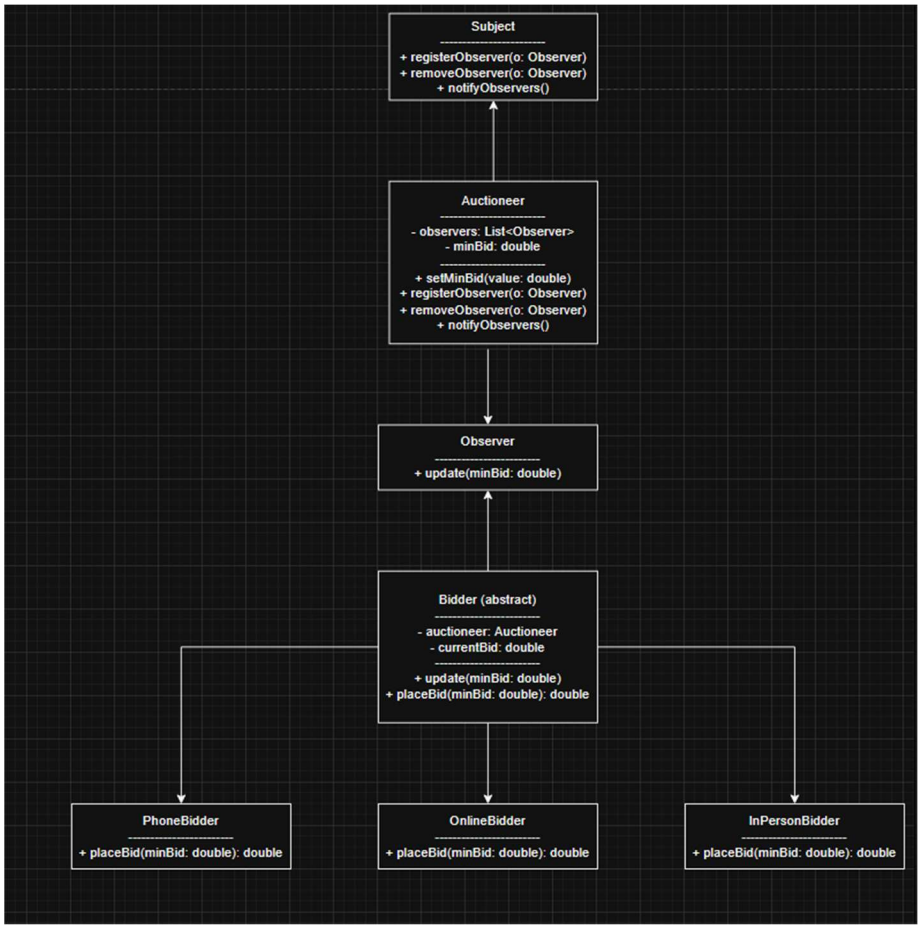
Problem 1:

Part A: (this is within AuctionSystem)

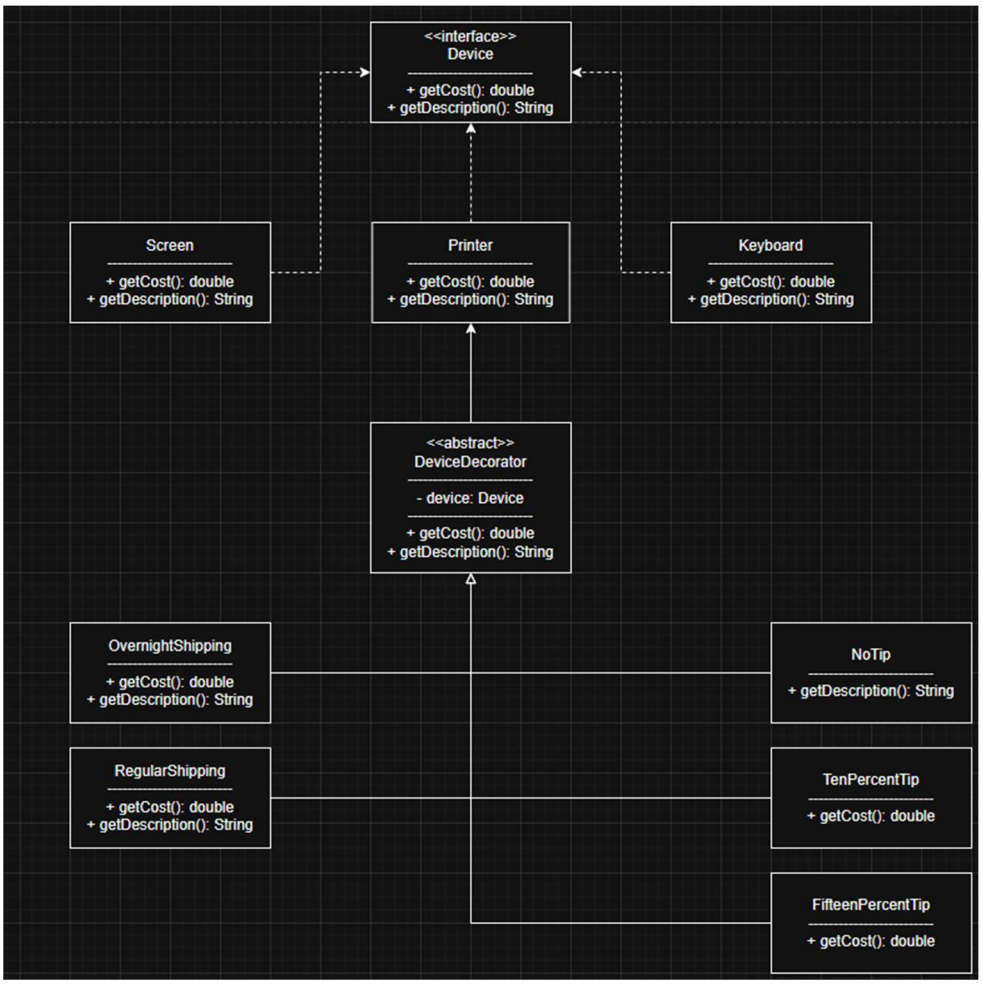


Part B:

Code is attached with PDF

Problem 2:

Part A:



Part B:

Code is attached

Problem 3:

A. Text Formatting in Word Processors

The Decorator pattern is most suitable because text formatting is layered and dynamic. Features like bold, italics, and underline can be added or removed independently while preserving the base text. Each formatting option acts as a wrapper that enhances the core text object without requiring many subclasses.

B. GPS Route Planning

The Strategy pattern fits best because route calculation requires selecting between interchangeable algorithms such as fastest, shortest, or scenic routes. Each routing method can be implemented as a separate strategy and chosen dynamically at runtime without changing the core navigation logic. This keeps the system flexible and modular.

C. Stock Market Alerts

The Observer pattern is ideal because stock data acts as a subject that notifies multiple observers (subscribers) when changes occur. Investors or apps can subscribe or unsubscribe dynamically, and all observers are automatically notified when stock prices update. This supports scalable, event-driven communication.